APPM 5460 Project Proposal

Chuan Wei Heng

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1 Background

This project is motivated by my ongoing research in Professor Ankur Gupta's laboratory, where I investigate the theoretical modeling of coupled multicomponent ion transport in electrolyte systems. In particular, I consider an aqueous electrolyte containing hydrogen (H^+), hydroxide (OH^-), sodium (Na^+), and chloride (Cl^-) ions.

The transport of these species is governed by diffusion and electromigration, with the ions coupled through the electrostatic potential generated by their charge distribution. In addition, hydrogen and hydroxide ions undergo an acid–base neutralization reaction to form water. The resulting system is described by the Poisson–Nernst–Planck equations coupled to chemical reaction kinetics.

Governing Equations

The conservation equation for each ionic species is

$$\frac{\partial c_i}{\partial t} = -\nabla \cdot \boldsymbol{J}_i + R_i, \qquad i \in \{H^+, OH^-, Na^+, Cl^-\}, \tag{1}$$

where c_i denotes the concentration, $\boldsymbol{J}_i$ the molar flux, and R_i the net rate of production of species i.

The ionic flux is described by the Nernst–Planck equation,

$$\boldsymbol{J}_i = -D_i \left(\nabla c_i + \frac{z_i F}{RT} c_i \nabla \phi \right), \tag{2}$$

where D_i and z_i are the diffusivity and valence of species i, respectively. Furthermore, F is Faraday's constant, R is the ideal gas constant, T is the absolute temperature, and ϕ is the electrostatic potential.

The acid–base reaction is modeled using the following second-order reaction kinetics:

$$R_{H^+} = R_{OH^-} = -k c_{H^+} c_{OH^-}, \tag{3}$$

$$R_{Na^+} = R_{Cl^-} = 0. \tag{4}$$

Finally, the electrostatic potential is determined self-consistently through Poisson's equation,

$$-\varepsilon \nabla^2 \phi = \rho_e = F \sum_i z_i c_i, \tag{5}$$

where ε denotes the permittivity and ρ_e is the local charge density.

Domain and Boundary Conditions

We consider a one-dimensional domain of length L, with prescribed ionic concentrations and electrostatic potentials at both boundaries.

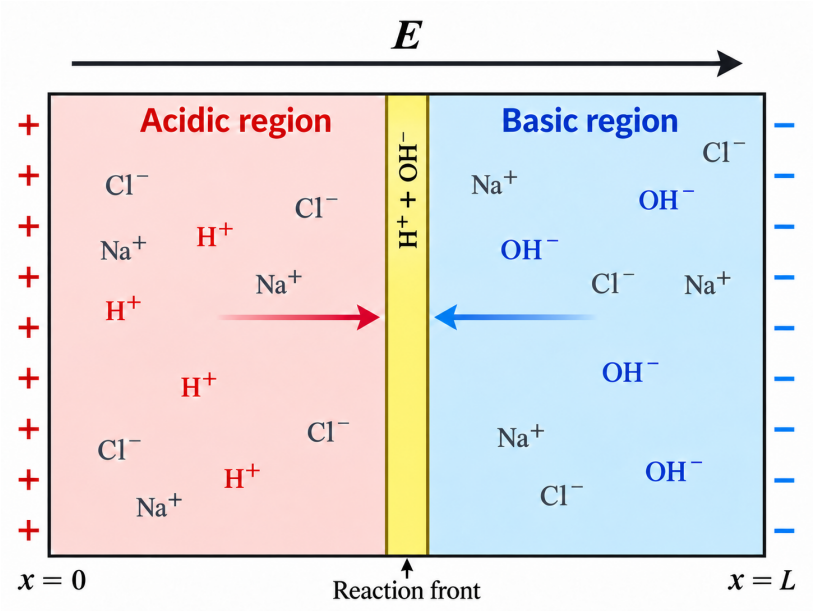


Figure 1: Schematic of the one-dimensional electrolyte domain and acid–base reaction front.

At the left boundary, $x = 0$,

$$\begin{aligned} c_{H^+} &= c_0, & c_{Na^+} &= c_s, \\ c_{OH^-} &= 0, & c_{Cl^-} &= c_0 + c_s, \\ \phi &= \phi_0. \end{aligned} \tag{6}$$

At the right boundary, $x = L$,

$$\begin{aligned} c_{H^+} &= 0, & c_{Cl^-} &= 0, \\ c_{Na^+} &= c_0, & c_{OH^-} &= c_0, \\ \phi &= \phi_L. \end{aligned} \tag{7}$$

The initial ionic concentrations in the interior of the domain are prescribed as

$$c_i(x, 0) = 0, \qquad 0 < x < L, \qquad \forall i. \tag{8}$$

2 Electroosmotic Instabilities

Our recent research has led us to hypothesize that electroconvective vortices may develop near an acid–base reaction front under an externally applied electric potential. This hypothesis is motivated by earlier work on electroconvective instabilities near ion-selective membranes, particularly the work of Rubinstein and Zaltzman [1].

In these systems, electrokinetic instabilities can generate vortical fluid motion near the membrane interface. This phenomenon is commonly associated with the Rubinstein–Zaltzman instability.

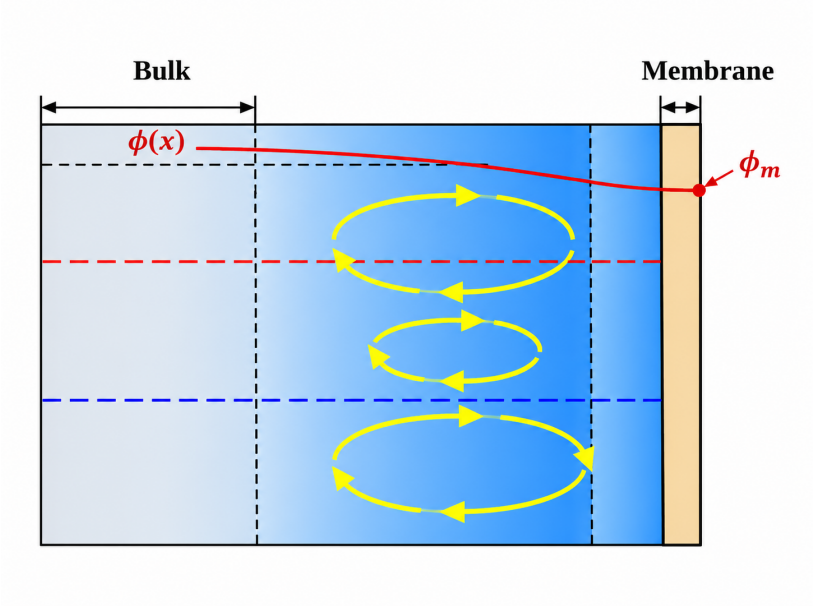


Figure 2: Schematic of electroconvective vortices near the interface of an ion-selective membrane.

Inspired by this mechanism, we seek to investigate whether a related instability can arise in a membraneless electrolyte system, where the acid–base reaction front creates spatial variations in ionic concentrations and electrical properties.

Unlike membrane-based systems, the proposed instability would occur without a physical ion-selective interface. Consequently, the existence and underlying mechanism of such an instability remain to be established.

Our central research question is therefore: *Can a Rubinstein–Zaltzman-type electroconvective instability arise near a membraneless acid–base reaction front?*

3 Proposed Investigation

The objective of this project is to investigate the dynamical stability of the coupled ion-transport system and determine whether electroconvective instabilities can arise under an applied electric potential.

We propose to begin by examining the steady-state solutions of the one-dimensional Poisson–Nernst–Planck–reaction system, which will serve as the base state for our stability analysis. Particular attention will be given to the spatial distributions of ionic concentrations, electric potential, and charge density near the reaction front.

We will then consider extending the model to incorporate fluid motion and investigate the response of the base state to small perturbations. By linearizing the governing equations about the steady-state solution, we aim to examine whether these perturbations grow or decay with time.

The primary mathematical objective is to identify the conditions under which the steady-state solution becomes unstable. Where feasible, we will investigate how this behavior depends on the imposed electric potential, reaction kinetics, and ionic transport properties.

If the full coupled stability analysis proves too extensive for the scope of the course, we will focus on a suitably simplified model that retains the essential coupling between ion transport, electrostatic forces, and fluid motion.

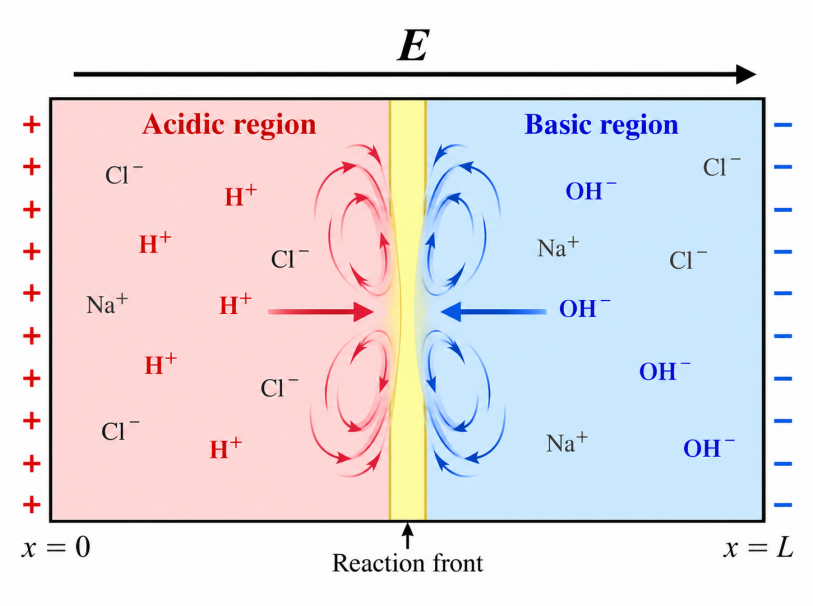


Figure 3: Schematic of hypothesized electroconvective vortices near a membraneless acid–base reaction front under an applied electric potential.

References

[1] Isaak Rubinstein and Boris Zaltzman. "Electro-osmotically induced convection at a permselective membrane". In: *Physical Review E* 62.2 (2000), p. 2238.